Excretory Products & their Elimination

SHORT NOTES

EXCRETORY WASTE

Metabolism — Excess ingestion

Results in accumulation of

- **Nitrogenous Wastes** – NH_3, urea, uric acid
- Other contents – CO_2, H_2O, ions (Na^+, K^+, Cl^-, PO_4^{3-}, SO_4^{2-})

Removed completely/Partially

Common Nitrogenous Wastes

❖ Nature of nitrogenous waste formed and their excretion vary among animals depending on the **habitat/availability of water**.

Common Nitrogenous Waste	Nature & Examples	Toxicity Level and Water Required	Special Features
Ammonia (Ammonia converts into urea in liver)	**Ammonotelic** ❖ Aquatic insects ❖ Many bony fishes ❖ Aquatic amphibians	Maximum	❖ Diffusion through gills surface or body surface as ammoium (NH_4^+) ions
Urea	**Ureotelic** ❖ Marine fishes ❖ Many terrestrial amphibians ❖ Mammals	Lesser	❖ **Kidneys** filter urea from blood
Uric acid	**Uricotelic** ❖ Land snails ❖ Insects ❖ Reptiles ❖ Birds	Least	❖ Pellet/Paste (Semi-solid) ❖ Minimum loss of water
❖ Kidneys do not play a significant role in removal of ammonia.			

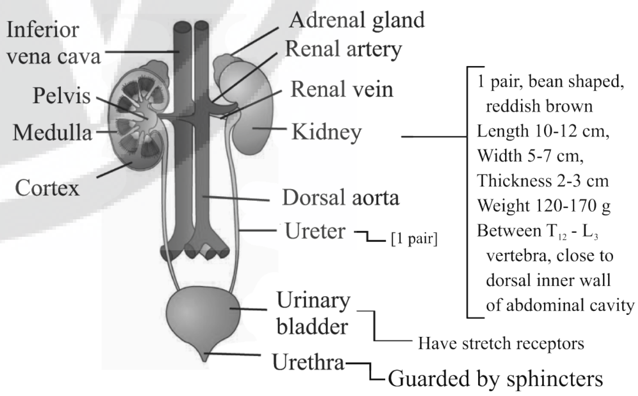
Different Excretory Structures

❖ Excretory strcuture eliminate nitrogenous waste & maintain fluid and ionic balance.

Structures	Examples
• Protonephridia/flame cells (osmoregulation)	• Platyhelminthes (*Planaria*) Rotifers Some annelids Cephalochordates (*Amphioxus*)
• Nephridia	• Annelids (Earthworms)
• Malpighian tubules	• Insects (Cockroaches)
• Antennal/Green glands	• Crustaceans (Prawn)

❖ Function of excretory structures:
- ✦ Eliminate nitrogenous wastes.
- ✦ Maintain ionic and acid-base balance of body fluids, i.e., osmoregulation.

HUMAN EXCRETORY SYSTEM



MICTURITION

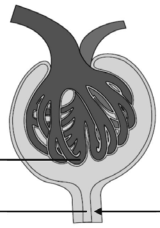
❖ Process of release of urine

❖ Mechanism -**Micturition reflex**

Urinary bladder (Store urine) $\xrightarrow{\text{Activates}}$ Stretch receptors $\xrightarrow{\text{signals}}$ CNS (voluntary signals) $\xrightarrow{\text{Send motor messages}}$ Urinary bladder

Smooth muscles of bladder contract and urethral sphincters relax $\xrightarrow{\text{Release}}$ Urine

URINE FORMATION



Non-Selective Process

1. **Glomerular filtration**

180 L filtrate/day

2. **Tubular reabsorption**

Nearly **99%** of filtrate get **reabsorbed** by renal tubules

Active process	Passive process
❖ Glucose, Na^+, amino acid	❖ Nitrogenous wastes, H_2O

3. **Tubular secretion**

❖ H^+, K^+ and ammonia secreted into filtrate
❖ Maintains ionic and acid base balance

Urine
1-1.5 L/day

- Filtration is due to pressure in the glomerular capillaries.
- Glomerular filtration rate (GFR) = Filtration/min 125 ml/min

REGULATION OF KIDNEY FUNCTION

Hypothalamus	**JGA/Renin-Angiotensin Mechanism**	**Heart**
❖ Works when GFR low, blood volume low. ❖ Osmoreceptors in hypothalamus activated. ❖ Release of ADH/Vasopressin ↓ Prevent diuresis	❖ Low GFR/Glomerular blood flow/ Glomerular blood pressure Activate↓ JG cells to release renin ↓ Angiotensinogen (Liver) → Angiotensin I ↓ ← ACE-Angiotensin converting enzyme **Angiotensin II** → Vasoconstriction ↓ ❖ Adrenal cortex aldosterone ↓ Reabsorption of Na^+ & water from distal part of tubule ↓ Blood pressure and GFR increase	❖ Increase blood flow to atria of heart ↓ Release of ANF (Atrial natriuretic factor) ↓ Vasodilation ↓ ❖ Blood pressure decrease ❖ Checks on Renin-Angiotensin Mechanism

Characteristics and Composition of Urine

- Colour - Light yellow
- pH = 6
- Odour - Characteristic
- Urea - 25-30 gm/day
- Glucosuria [Glucose in urine] / Ketonuria [Ketone in urine] } Diabetes mellitus

Role of Other Organs in Excretion

Accessory Structure	Function
Lungs	• Remove large amount of CO_2. • Approximately 200 mL/min. • Remove significant quantity of water.
Liver (Largest gland)	• Remove bile-containing substances along with digestive wastes.
Skin • Sweat gland	• Removes NaCl, urea and lactic acid. • Facilitates cooling effect.
• Sebaceous gland	• Removes sterols, hydrocarbons, waxes.
Salivary glands	• Small amount of nitrogenous wastes are eliminated through saliva.

Disorders of Excretory System

Disorders	Symptoms or Treatment
Renal calculi	Stone or insoluble mass of crystalised salts (e.g., oxalates).
Glomerulonephritis	Inflammation of glomeruli of kidney.
Renal/kidney failure	Malfunctioning of kidneys may lead to kidney failure. **Treatment** (*i*) Haemodialysis: Process to remove urea from blood. • Boon for thousands of uremic patients all over the world. • Composition of dialysing fluid is same as plasma except the nitrogenous wastes.
	(*ii*) Kidney transplantation • Ultimate method in correction of acute renal failure. • Functional kidney is taken from donor. • To minimise rejection, close relatives are preferred as donor. • Modern clinical problems have increased success rate of such complicated techniques.